表 1: 层间边际贡献—ΔAUROC 与统计显著性

转移	加入的组件	ΔAUROC	DeLong z	p-value	ΔCohen_d	ΔBrier
L0 → L1	+ PBC Detection	+0.2141	6.939	0.0000	+2.2097	−0.0736
L1 → L2	+ Evidence + Identity	+0.0094	0.677	0.4987 —	+0.1075	−0.0156
L2 → L3	+ Observer (Full System)	−0.0000	-0.000	1.0000 —	+0.1959	−0.0093